

Healthcare Software Modernization with AI-Assisted Authorization Workflow

How Technijian modernized a healthcare practice's live web application, stood up an MCP-compatible data service, and integrated an OpenAI Assistant to auto-generate authorization checklists — all inside a HIPAA-aware development cycle.

INDUSTRY	ENVIRONMENT	SERVICES ENGAGED	ENGAGEMENT SCOPE
Healthcare / Medical Practice	Live web app + DB + AI	Custom Dev, MCP, AI Integration	70+ hrs development

THE CHALLENGE

A medical practice ran a live web application supporting day-to-day clinical and authorization workflows. The **user interface was aging**, the **database schema needed to evolve** to support new data requirements, and clinical staff were spending too much time manually assembling **authorization checklists** — a task that was repetitive, error-prone, and a drag on patient throughput.

The practice also wanted to explore **AI assistance** for the checklist task, but without risking data-privacy violations or shipping something untested into a clinical workflow. They needed a partner who could do the software modernization *and* the AI integration inside a single engagement with HIPAA-aware controls.

- ✓ **Aging UI** on a live healthcare web application
- ✓ **Database schema limits** blocking new data-capture requirements
- ✓ **Manual authorization checklists** — repetitive, error-prone, slow
- ✓ **No programmatic data service** between UI and downstream systems
- ✓ **AI aspiration without AI governance** — interest in OpenAI, zero tooling in place
- ✓ **HIPAA posture** must be preserved through every change

THE SOLUTION

Technijian delivered the work in a **three-track development engagement**: UI + database modernization of the live web application, an MCP-compatible data service stood up to feed the UI, and an OpenAI Assistant integrated to auto-generate authorization checklists formatted to UI specifications.

TRACK 1

UI & Database Update

Iterative UI redesign for accessibility with user feedback loops; database schema updates to support new data requirements; weekly stakeholder check-ins and change logs.

TRACK 2

MCP Data Service

Stood up an MCP-compatible server to facilitate clean data access between UI and database, with monitoring for uptime and performance metrics.

TRACK 3

OpenAI Assistant

Integrated an OpenAI Assistant to generate authorization checklists formatted to UI specifications, with defined input data formats and data-privacy compliance.

TECHNICAL & PROCESS HIGHLIGHTS

- ✓ **Iterative delivery** with user feedback every cycle — not a waterfall reveal
- ✓ **Weekly check-ins** with change logs for every UI and DB change
- ✓ **MCP-compatible architecture** — future-proof for AI agent access patterns
- ✓ **Server monitoring** for MCP uptime and performance from day one
- ✓ **Defined input formats** + data-privacy compliance at the AI boundary

TECHNOLOGY STACK

- Web App (UI modernization)
- SQL / relational DB
- MCP Server
- OpenAI Assistants API
- Monitoring & uptime
- HIPAA-aware dev cycle

WHY IT MATTERS

- Faster authorization workflow
- Fewer manual errors
- Governed AI, not rogue AI

ENGAGEMENT SCALE

3 PARALLEL TRACKS	70+ DEV HOURS	1 MCP SERVICE LAUNCHED	AI ASSISTANT INTEGRATED
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Modernize the app, add AI, keep HIPAA posture. One partner.

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THE RESULTS

The practice came out of the engagement with a **modernized UI**, a **schema that fits new data requirements**, an **MCP-compatible data service** the team can build on, and an **OpenAI Assistant that generates authorization checklists** in the exact format the UI expects — removing a repetitive clinical-admin task from the human queue.

OUTCOME METRIC	RESULT
UI modernization	Live web application UI updated for better accessibility; iterative delivery with user feedback incorporated every cycle — not a waterfall reveal.
Database schema	Schemas updated to incorporate new data requirements that the previous model couldn't represent — unblocking downstream features.
MCP data service	New MCP-compatible server stood up to facilitate clean data access between UI and database; monitoring in place for uptime and performance metrics.
OpenAI Assistant integration	AI assistant generating authorization checklists formatted to UI specifications, with defined input data formats and governed data flow.
Authorization throughput	Repetitive manual checklist assembly shifted from human time to AI-assisted generation with human review — freeing clinical-admin capacity.
AI governance	Compliance-with-data-privacy maintained as a design constraint, not an afterthought — defined input formats, monitored performance, human-in-the-loop validation.
Delivery model	Iterative agile cycles with weekly stakeholder check-ins and formal change logs — every UI, DB, and AI change traceable.

REPRESENTATIVE DELIVERABLES

UI redesign — Modernized the live web application's user interface for better accessibility; iterative approach with user feedback shaping each cycle.

MCP-compatible server — Stood up the MCP data service to facilitate data access for the updated application, with monitoring for uptime and performance.

Weekly change logs — Maintained written change logs for every UI, DB, and integration change — ready for compliance review and knowledge transfer.

Database schema update — Updated schemas to support the practice's new data requirements, unblocking UI and reporting features the old model couldn't express.

OpenAI Assistant — Built the assistant to generate authorization checklists formatted to UI specs, with defined input schemas and governed data flow.

Monitoring & uptime tooling — Established server monitoring for the MCP data service, not left as a post-launch "we'll get to it" item.

REPRESENTATIVE OUTCOME

"A healthcare practice modernized its live web application, stood up a future-proof MCP data service, and integrated an OpenAI Assistant to automate authorization checklists — delivered as one engagement, under iterative agile cycles, with HIPAA-aware controls preserved through every change."

— Anonymized engagement summary · based on 70-hour custom software development proposal